

Induced EMF of a Dropped Magnet

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1 Objective

We drop a magnet through a coil from ten different heights and read the peak-to-peak voltage V_{peak} off the scope each time. Then we plot V_{peak} against drop height h and against the magnet's speed at the coil $v = \sqrt{2gh}$.

2 Theory

Faraday's law gives the voltage induced across a coil of N turns by a changing flux Φ_B through one turn,

$$\varepsilon = -N \frac{d\Phi_B}{dt}. \quad (1)$$

For a magnet falling along the coil axis, the flux changes faster when the magnet moves faster, so the induced voltage is proportional to the magnet's speed v at the coil.

In free fall from release height h above the coil, $v = \sqrt{2gh}$, so $V_{\text{peak}} \propto v \propto \sqrt{h}$: V_{peak} vs. v should be linear through the origin, and V_{peak} vs. h should follow a square-root curve. The biggest source of uncertainty is where we place the magnet at release, which we take as ± 0.5 cm.

3 Apparatus

- Aluminum support rod (~ 1 m) C-clamped to the bench; a fixture at the top anchors a vertical guide line through the coil bore so the magnet falls along the coil axis.
- Small induction coil of insulated magnet wire on a bracket mounted to the rod at bench-top height, axis vertical, leads to scope ch 1 via a BNC adapter.
- Small permanent drop magnet; same magnet, same pole down each drop.
- Oscilloscope (Tektronix TDS 1002B, 60 MHz).
- Wooden meterstick clamped to the rod, reading h ; $\delta h = \pm 0.5$ cm from where we place the magnet.

Circuit schematic

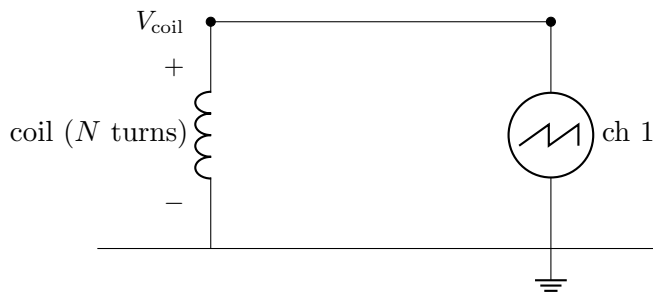


Figure 1: Coil leads connected to oscilloscope ch 1.

4 Procedure

1. Wire the coil to scope ch 1; set the trigger to Edge / CH1 / Rising / Normal / AC, and the time-base to 5 ms/div so the induced pulse fits on screen across the full height range (see Calculations for the $v = \sqrt{2gh}$ and x/v estimate).
2. Clamp the meterstick vertically against the support rod with its zero at the coil top; use that reference for all ten trials.
3. Pick ten drop heights spanning roughly 10 cm to 100 cm. For each one: line up the bottom of the magnet at the target height, record $h \pm 0.5$ cm, let it go without pushing or spinning it, and read the peak-to-peak voltage $V_{\text{peak}} = V_{\text{max}} - V_{\text{min}}$ straight off the scope display.

If a drop tumbles or the trace runs off the screen, re-take that height.

5 Data

We dropped the magnet from ten different heights and read $V_{\text{peak}} = V_{\text{max}} - V_{\text{min}}$ straight off the scope each time. Table 1 has one row per drop. Heights ranged from $h \approx 0.20$ m up to about 0.96 m, capped by the guide-line clearance above the coil. We take $\delta h = \pm 0.5$ cm from where we place the magnet, and $\delta V_{\text{peak}} = \pm 0.04$ V from reading the scope at the V/div setting we used. Figure 2 shows the data with a $\sqrt{h}$ curve drawn on top.

Trial	h (m)	V_{peak} (V)
1	0.20 ± 0.005	1.12 ± 0.04
2	0.30 ± 0.005	1.20 ± 0.04
3	0.35 ± 0.005	1.28 ± 0.04
4	0.40 ± 0.005	1.36 ± 0.04
5	0.47 ± 0.005	1.40 ± 0.04
6	0.55 ± 0.005	1.44 ± 0.04
7	0.65 ± 0.005	1.52 ± 0.04
8	0.80 ± 0.005	1.84 ± 0.04
9	0.87 ± 0.005	1.92 ± 0.04
10	0.96 ± 0.005	2.12 ± 0.04

Table 1: Drop height h and peak-to-peak coil voltage V_{peak} for each of the ten drops.

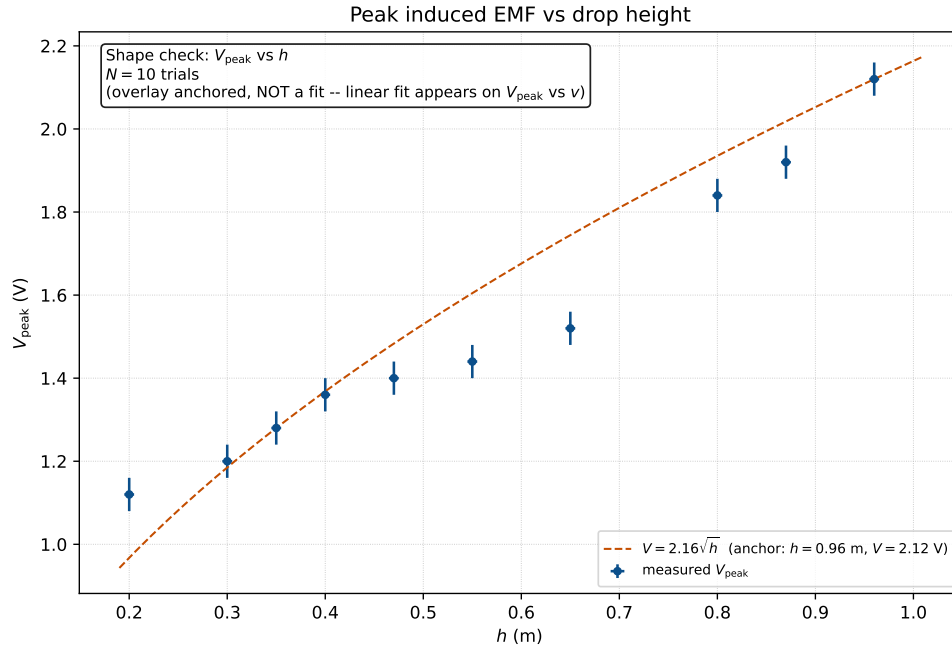


Figure 2: V_{peak} vs. drop height h , with $\sqrt{h}$ overlay.

6 Calculations and Uncertainty

Propagation rules

$$\delta(A \pm B) = \sqrt{\delta_A^2 + \delta_B^2}$$

$$\frac{\delta(AB)}{|AB|} = \frac{\delta(A/B)}{|A/B|} = \sqrt{\left(\frac{\delta_A}{A}\right)^2 + \left(\frac{\delta_B}{B}\right)^2}$$

$$\delta f = \left| \frac{df}{dx} \right| \delta x$$

Time-base estimate (worst case, $h = 0.96$ m)

Before sweeping the heights, we picked the scope's time-base from a quick energy estimate. Energy conservation gives the magnet's speed at the coil, and dividing the solenoid's axial length by that speed gives roughly how long the induced pulse should last; the shortest pulse sits at the largest drop height, so that sets the upper bound on the time-base.

$$mgh = \frac{1}{2}mv^2 \implies v = \sqrt{2gh}$$

$$v = \sqrt{2 \cdot 9.81 \cdot 0.96} = 4.34 \text{ m s}^{-1}$$

$$t = \frac{x}{v} = \frac{0.04}{4.34} \approx \boxed{9.2 \text{ ms}}$$

We set the scope to 5 ms/div: the worst-case transit spans roughly two divisions, and lower drop heights span a few more, so the full bipolar $V(t)$ feature fits on the ten-division horizontal screen across the whole sweep.

Velocity at the coil (Trial 6)

Inputs: $h = 0.55 \pm 0.005$ m, $g = 9.81$ m s⁻².

$$v = \sqrt{2gh}, \quad \delta v = \sqrt{\frac{g}{2h}} \delta h$$

$$v = \sqrt{2 \cdot 9.81 \cdot 0.55} = 3.285$$

$$\delta v = \sqrt{\frac{9.81}{2(0.55)}} \cdot 0.005 = 0.015$$

$$v = \boxed{3.29 \pm 0.02 \text{ m s}^{-1}}$$

We ran this same propagation for every row of Table 1 to get the $v \pm \delta v$ values that go into the regression below.

Linear regression: $V_{\text{peak}} = m v + b$

We fit a least-squares line to V_{peak} versus $v = \sqrt{2gh}$ across all ten trials (Fig. 3):

$$m = \boxed{0.410 \pm 0.039 \text{ V/(m/s)}}$$

$$b = \boxed{0.203 \pm 0.128 \text{ V}}$$

$$R^2 = 0.933, \quad N = 10$$

The fit equation, R^2 , and N are annotated on the figure itself.

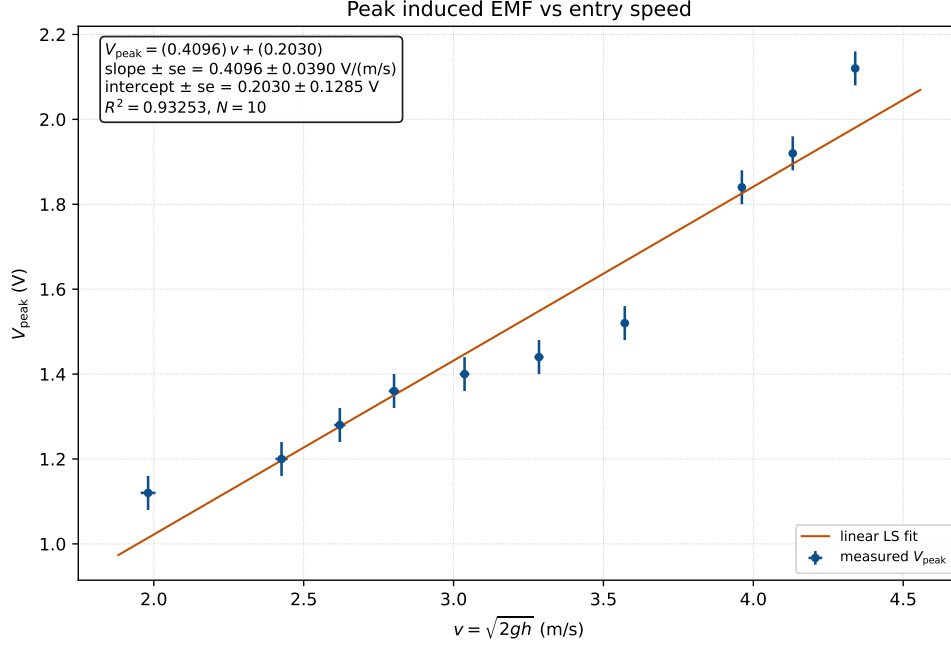


Figure 3: V_{peak} vs. coil-entry speed $v = \sqrt{2gh}$, with linear fit.

t' test on the intercept

Faraday's law says the line should go through the origin, so we expect $b = 0$.

$$t' = \frac{|b - 0|}{\delta b}$$

$$= \frac{0.203}{0.128}$$

$$t' = \boxed{1.58}$$

That puts t' in the borderline band ($1 \leq t' \leq 3$): not quite consistent with zero, but not in outright disagreement either. So from this dataset we can't pin down whether the intercept is really zero.

7 Conclusion

Over ten drops, the peak-to-peak coil voltage grew linearly with the magnet's speed at the coil $v = \sqrt{2gh}$, with slope 0.41 ± 0.04 V/(m/s) and $R^2 = 0.93$. This is basically what Faraday's law predicts: a faster magnet at the coil gives a bigger induced voltage. By eye the fit line passes through or near most of the error bars; the trial at $h \approx 0.65$ m sits visibly below the line and the highest- h point sits above it, which is where most of the residual scatter comes from. The fitted intercept 0.20 ± 0.13 V is small but not quite zero ($t' = 1.58$); since it's only about a tenth of the V_{peak} range we swept, we read it as scope reading error rather than a real offset. The biggest source of uncertainty is the ± 0.5 cm placement of the magnet against the meterstick.